

Five Number Summary & Box Plot

Tutorial 7- How To Construct A Box Plot For Outliers

1. Introduction: The Danger of Outliers

Hello students! Welcome to the intermediate level of Statistics. Today we are learning how to catch a hidden "thief" in our data. This thief is called an **Outlier**.

Outlier: A data point that is completely different from the rest of the distribution. It can be a weirdly small number or a weirdly massive number.

Why do we care about outliers? Because when we use Machine Learning (specifically Feature Engineering), outliers can confuse our computer models and give us completely wrong predictions. Detecting and handling them is one of the most important jobs of a Data Scientist!

Real-Life Example:

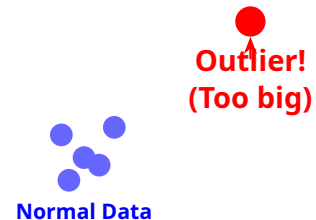
Imagine measuring the ages of students in an 8th-grade classroom. You get: 13, 14, 13, 14, 15... and suddenly, you see an age of **87**!

That 87 is an **Outlier**. It doesn't belong with the normal student group.

Practice Question:

Q: Look at this data: 10, 11, 12, 11, 10, 1000. Which number is the outlier?

A: 1000 is the outlier because it is completely different from the normal range of 10-12.



Topic Summary: Outliers

Outliers are extreme values that don't fit the rest of the data. They cause big problems in Machine Learning, so we must learn how to detect them.

2. The Five Number Summary

To build a statistical tool that catches outliers, we first need 5 magical numbers from our dataset. Together, these are called the **Five Number Summary**.

Important Note & Extra Info:

The 5 Magical Components:

1. **Minimum:** The smallest normal value in the dataset.
2. **First Quartile (Q1):** The 25th Percentile (the middle of the first half).
3. **Median (Q2):** The 50th Percentile (the exact center).
4. **Third Quartile (Q3):** The 75th Percentile (the middle of the second half).
5. **Maximum:** The largest normal value (excluding outliers).

Once we find these 5 numbers, we use them to draw a special chart called a **Box Plot**. The main goal of a Box Plot is to instantly show us where the outliers are hiding!

5-Number Summary

1. Minimum
2. Q1 (25%)
3. Median
4. Q3 (75%)
5. Maximum

3. How to Detect Outliers (The Fence Method)

To legally declare a number as an "Outlier", we use math to build two invisible walls called **Fences**. Anything outside these fences is captured as an outlier!

Important Note & Extra Info:

Step 1: Find the IQR (Interquartile Range)

$$\text{IQR} = \text{Q3} - \text{Q1}$$

Step 2: Build the Lower Fence

$$\text{LowerFence} = \text{Q1} - 1.5 \times \text{IQR}$$

Any number smaller than this fence is an Outlier.

Step 3: Build the Higher (Upper) Fence

$$\text{HigherFence} = \text{Q3} + 1.5 \times \text{IQR}$$

Any number bigger than this fence is an Outlier.



4. Let's Calculate! (A Real Example)

Real-Life Example:

Our Dataset: 1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, **27**

Total numbers (N) = 19. Is 27 an outlier? Let's prove it with math!

1. **Find Q1 (25th Percentile):** Index = $(25/100) \times (19 + 1) = 0.25 \times 20 =$ **5th Index**.

Looking at the 5th number in our list: **Q1 = 3**.

2. **Find Q3 (75th Percentile):** $\text{Index} = (75/100) \times (19 + 1) = 0.75 \times 20 = \mathbf{15th}$ Index.
Looking at the 15th number in our list: **Q3 = 7.**
 3. **Find IQR:** $IQR = Q3 - Q1 = 7 - 3 = \mathbf{4}.$
 4. **Lower Fence:** $Q1 - (1.5 \times IQR) = 3 - (1.5 \times 4) = 3 - 6 = \mathbf{-3}.$
 5. **Higher Fence:** $Q3 + (1.5 \times IQR) = 7 + (1.5 \times 4) = 7 + 6 = \mathbf{13}.$
- The Conclusion:** Our "Safe Zone" is from **-3 to 13**. Is there any number bigger than 13? YES!
27 is much bigger than 13. Therefore, **27 is officially an Outlier!**

5. Creating the Box Plot

Now that we caught the outlier (27), we can define our final Five-Number Summary to draw the Box Plot:

- **Min:** 1
- **Q1:** 3
- **Median:** 5 (*The 10th index*)
- **Q3:** 7
- **Max:** 9 (*Notice: We use the biggest normal number, not the outlier!*)

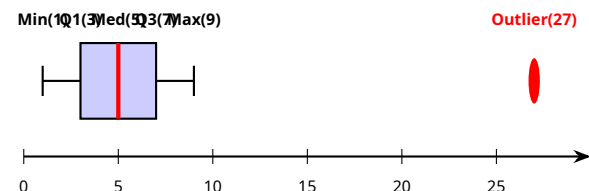
How to draw it: 1. Draw a box stretching from Q1 (3) to Q3 (7).

2. Draw a line inside the box at the Median (5).

3. Draw "whiskers" (lines) extending to Min (1) and Max (9).

4. Place a small dot far away for the Outlier (27).

Tip: In Python, you can use the Seaborn library to draw this automatically with just one line of code!



Overall Revision Summary

- **Outliers:** Dangerous extreme values in data that can mess up Machine Learning predictions.
- **Five Number Summary:** The 5 core values needed to understand data distribution (Min, Q1, Median, Q3, Max).
- **Detecting Outliers Formula:**
 - $IQR = Q3 - Q1$
 - Lower Fence = $Q1 - 1.5 \times IQR$
 - Higher Fence = $Q3 + 1.5 \times IQR$
 - Anything outside these fences is an Outlier!
- **Box Plot:** A visual chart created using the Five-Number Summary. The box is made from Q1 to Q3. The whiskers reach the Min and Max. Outliers are plotted as individual dots!

Keep practicing! Finding outliers is a superpower for every Data Scientist. In the next video, we will learn how to code this in Python!